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Studies on Electrophilic Interaction of 2,3-Allenols with Electrophilic Halogen Reagents: Selective Synthesis of 2,5-Dihydrofurans, 3-Halo-3-alkenals, or 2-Halo-2-alkenyl Ketones

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Abstract: The reaction of primary 2,3-allenols with iodine (I_2) afforded 2,5-dihydrofurans while that of readily available 1-aryl or 1-methyl substituted 2,3-allenols with bromine (Br_2), *N*-bromosuccinimide (NBS), I_2 or *N*-iodosuccinimide (NIS) formed the not easily available but synthetically useful 3-halo-3-alkenals and 2-halo-2-alkenyl ketones with good selectivity and yields *via* a sequential electrophilic interaction of X^+ with the allene moiety, 1,2-aryl or 1,2-proton shift, and H^+ elimination process.

Keywords: alcohols; aldehydes; allenes; electrophilic addition; halogens; ketones

Introduction

Elecotrophilic additions of alkenes and alkynes with "X^+" have provided many useful methodologies for the efficient synthesis of halides.[1] The electrophilic addition of allenes with "X^+" is also quite attractive since 2-haloallylic halides or alcohols are usually formed.[2] The electrophilic addition of alkyl-substituted allenes afforded terminal- or center-attack products, depending on the structures of the allenes and electrophiles:[3] Chlorination of propadiene afforded a mixture of 2,3-dichloro-1-propene and propargylic chloride[4] while that of 3-methyl-1,2-butadiene afforded a mixture of regioisomeric 2,3-dichloro-3-methyl-1-butene and 1,2-dichloro-3-methyl-2-butene.[5] The manganese-mediated chlorination of monosubstituted allenes afforded a mixture of 2,3-dichloro-1-alkenes and 1,2-dichloro-2-alkenes.[6] Bromination of phenyl-1,2-propadiene in MeOH at 0 °C also led to a regioisomeric mixture of 2-bromoallylic methyl ethers, however, in a non-polar solvent such as CS_2, CH_2Cl_2, or CCl_4 only 2,3-dibromo-1-phenyl-1-propene was formed.[7] Iodination of dialkyl-substituted allenes formed 1,2-diiodo-2-alkenes in $CHCl_3$ or CH_2Cl_2 whereas in the presence of $Hg(OAc)_2$ in MeOH the reaction afforded a regioisomeric mixture of 2-iodoallylic ethers.[8] In our laboratory we have obsevered that the reaction of "X^+" with functionalized allenes afforded cyclic alkenyl halides[9] or the halohydroxylation products.[10] In 1993, Friesen et al. reported that the reaction of 2,3-allenols with I_2 in Et_2O afforded 3,4-diiodo-2-alken-1-ol with a stereoselectivity of 7:1 to 9:1 (*Z*/*E*).[11] In a primary communication we have reported that the sequential electrophilic addition reaction with "X^+" and 1,2-migration of 2,3-allenols afforded 3-halo-3-enals.[12] In this paper, we report our detailed study in this area.

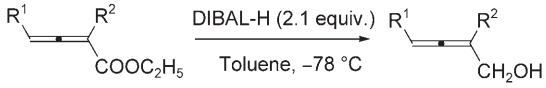
Results and Discussion

Preparation of the Starting Materials 2a–i

Primary 2,3-allenols **2a–i** used in this study were prepared by reduction of the related 2,3-allenates **1a–i** with DIBAL-H (Table 1).[13]

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Table 1. Synthesis of primary 2,3-allenols **2a–i**.



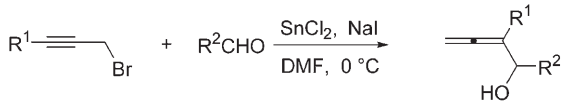
1 2

Entry	R^1	R^2	Time [h]	Yield [%] of **2a–i**
1	n-C_7H_{15}	H (**1a**)	5.3	23 (**2a**)
2	n-C_6H_{13}	CH_3 (**1b**)	2.5	36 (**2b**)
3	n-C_7H_{15}	CH_3 (**1c**)	3.8	57 (**2c**)
4	n-C_6H_{13}	n-C_3H_7 (**1d**)	6.2	70 (**2d**)
5	n-C_7H_{15}	n-C_3H_7 (**1e**)	6.7	53 (**2e**)
6	CH_3	Bn (**1f**)	3.5	32 (**2f**)
7	n-C_3H_7	Bn (**1g**)	4.8	33 (**2g**)
8	n-C_4H_9	Bn (**1h**)	3.3	38 (**2h**)
9	n-C_6H_{13}	Bn (**1i**)	3.5	40 (**2i**)

Preparation of Secondary 2,3-Allenols 5a–q

Secondary 2,3-allenols **5a–q** were prepared from the reaction of the corresponding propargylic bromides with aldehydes in the presence of NaI and $SnCl_2$ (Table 2).[14]

Table 2. Synthesis of secondary 2,3-allenols **5a–q**.

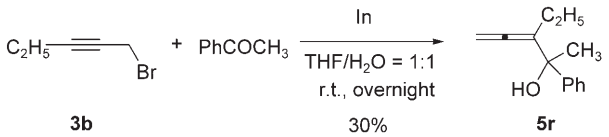


3 4 5

Entry	R^1	R^2	Time [h]	Yield [%]
1	n-C_4H_9	Ph	12	40 (**5a**)
2	n-C_4H_9	m,p-$OCH_2OC_6H_3$	15	45 (**5b**)
3	n-C_4H_9	p-$MeOC_6H_4$	10	45 (**5c**)
4	n-C_4H_9	p-MeC_6H_4	12	57 (**5d**)
5	n-C_2H_5	m,p-$OCH_2OC_6H_3$	16	44 (**5e**)
6	n-C_2H_5	p-$MeOC_6H_4$	12	66 (**5f**)
7	n-C_2H_5	p-MeC_6H_4	11	51 (**5g**)
8	allyl	m,p-$OCH_2OC_6H_3$	12	62 (**5h**)
9	allyl	p-$MeOC_6H_4$	11	41 (**5i**)
10	allyl	p-MeC_6H_4	17	56 (**5j**)
11	n-C_4H_9	α-Naphthyl	17	62 (**5k**)
12	n-C_4H_9	m-BrC_6H_4	20	58 (**5l**)
13	n-C_4H_9	p-ClC_6H_4	11	46 (**5m**)
14	n-C_4H_9	p-$NO_2C_6H_4$	17	66 (**5n**)
15	n-C_4H_9	o,p-$Cl_2C_6H_3$	20	49 (**5o**)
16	n-C_4H_9	o-ClC_6H_4	20	52 (**5p**)
17	n-C_4H_9	CH_3	10	46 (**5q**)

Preparation of Tertiary 2,3-Allenol 5r

The tertiary 2,3-allenol **5r** was prepared from the reaction of the corresponding propargylic bromide with acetophenone in the presence of In in $THF/H_2O = 1/1$ (Scheme 1).[15]



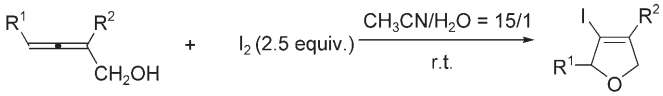
3b 30% 5r

Scheme 1.

Electrophilic Cyclization of Primary 2,3-Allenols 2 with I_2

Our initial study began with the reaction of 2-propyldeca-2,3-dien-1-ol **2d** with 2.5 equiv. of I_2 in $CH_3CN/H_2O = 15/1$ at room temperature.[12] Instead of the diiodination product,[11] 2,5-dihydrofuran **6d** was formed in 60% yield. Then, we studied the scope of this cyclization reaction with some of the typical results being summarized in Table 3. It can be concluded that differently substituted primary 2,3-allenols **2a–i** can be used to prepare the 2,4-disubstituted-3-iodo-2,5-dihydrofurans **6a–i** in moderate to good yields.

Table 3. Electrophilic cyclization of primary 2,3-allenols **2** with I_2.[a]



2 6

Entry	R^1	R^2	Time [h]	Isolated Yield [%] of **6**
1	n-C_7H_{15}	H (**2a**)	1.3	65 (**6a**)
2	n-C_6H_{13}	CH_3 (**2b**)	20	60 (**6b**)
3	n-C_7H_{15}	CH_3 (**2c**)	16	72 (**6c**)
4	n-C_6H_{13}	n-C_3H_7 (**2d**)	22	60 (**6d**)
5	n-C_7H_{15}	n-C_3H_7 (**2e**)	22	63 (**6e**)
6	CH_3	Bn (**2f**)	1.7	67 (**6f**)
7	n-C_3H_7	Bn (**2g**)	10	73 (**6g**)
8	n-C_4H_9	Bn (**2h**)	2.5	75 (**6h**)
9	n-C_6H_{13}	Bn (**2i**)	1.7	82 (**6i**)

[a] The reaction was carried out using 0.4–0.5 mmol of 2,3-allenols.

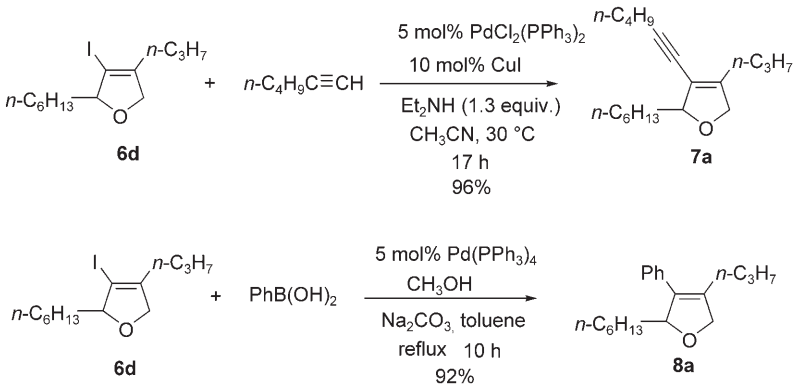
The carbon-iodine bond in **6d** may undergo a coupling reaction with 1-hexyne or phenylboronic acid to afford 3-hexynyl- or phenyl-substituted 2,5-dihydrofuran **7a** or **8a**, respectively, in excellent yields (Scheme 2).

Reaction of Secondary 2,3-Allenols with X_2

However, when we tried to extend this reaction to 1-aryl-substituted 2,3-allenols, the expected 2-aryl-2,5-dihydrofuran **9** was not formed. The reaction of 1-phenyl-2-butyl-2,3-butadienol **5a** with Br_2 in aqueous

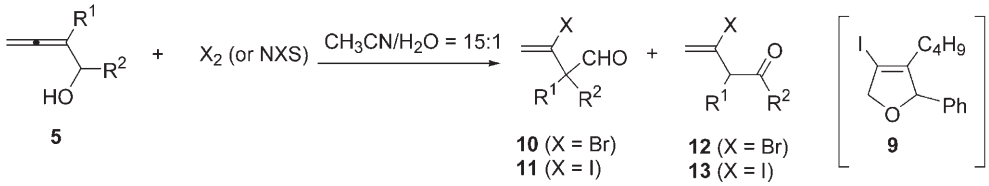
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Scheme 2.

Table 4. Reaction of secondary 2,3-allenols **5** with X_2 (or NBS).



Entry	R^1	R^2	X^+ [equiv]	Temp. [°C]	Time [min]	Yield [%] of **10** or **11**	**10**/**12** or **11**/**13**[a]
1	n-C_4H_9	Ph (**5a**)	Br_2 (2)	0	30	80 (**10a**)	≥95:5
2	n-C_4H_9	m,p-$OCH_2OC_6H_3$ (**5b**)	Br_2 (2)	r.t.	50	78 (**10b**)	98:2
3			NBS (1.2)	0	30	93 (**10b**)	99:1
4			I_2 (2)	r.t.	60	80 (**11b**)	>99:1
5	n-C_4H_9	p-$MeOC_6H_4$ (**5c**)	Br_2 (2)	r.t.	90	71 (**10c**)	≥99:1
6			NBS (1.2)	r.t.	30	88 (**10c**)	99:1
7			I_2 (2)	r.t.	30	77 (**11c**)	>99:1
8			NIS (2)	r.t.	30	89 (**11c**)	>99:1
9	n-C_4H_9	p-MeC_6H_4 (**5d**)	Br_2 (1.2)	0	60	87 (**10d**)	97:3
10			NBS (1.2)	0	140	90 (**10d**)	97:3
11			I_2 (2)	0	25	68 (**11d**)	97:3
12	C_2H_5	m,p-$OCH_2OC_6H_3$ (**5e**)	Br_2 (1.2)	0	25	78 (**10e**)	>99:1
13			NBS (1.2)	0	30	84 (**10e**)	>99:1
14			I_2 (2)	r.t.	30	69 (**11e**)	>99:1
15			NIS (2)	r.t.	25	87 (**11e**)	>99:1
16	C_2H_5	p-$MeOC_6H_4$ (**5f**)	Br_2 (1.2)	0	30	80 (**10f**)	>99:1
17			NBS (1.2)	0	40	88 (**10f**)	>99:1
18			I_2 (2)	0	30	83 (**11f**)	>99:1
19	C_2H_5	p-MeC_6H_4 (**5g**)	Br_2 (1.2)	0	30	86 (**10g**)	97:3
20			NBS (1.2)	0	35	64 (**10g**)	97:3
21			I_2 (2)	0	30	55 (**11g**)	97:3
22			NIS (1.2)	r.t.	40	79 (**11g**)	96:4
23	allyl	m,p-$OCH_2OC_6H_3$ (**5h**)	Br_2 (1.2)	0	30	68 (**10h**)	>99:1
24			NBS (1.2)	0	30	88 (**10h**)	>99:1
25			I_2 (1.2)	r.t.	50	complicated	-
26	allyl	p-$MeOC_6H_4$ (**5i**)	Br_2 (1.2)	0	100	57 (**10i**)	>99:1
27			NBS (1.2)	0	50	62 (**10i**)	>99:1
28			I_2 (1.2)	r.t.	30	complicated	-
29			I_2 (1.2)	−20	25	37 (**11i**)	≥98:2
30	allyl	p-MeC_6H_4 (**5j**)	Br_2 (1.2)	0	55	50 (**10j**)	≥97:3
31			NBS (1.2)	0	60	71 (**10j**)	≥97:3
32	n-C_4H_9	α-naphthyl (**5k**)	Br_2 (2)	0	30	73 (**10k**)	90:10
33			NBS (1.2)	0	50	76 (**10k**)	93:7
34	n-C_4H_9	m-BrC_6H_4 (**5l**)	Br_2 (1.2)	r.t.	30	49 (**10l**/**12l**)	80:20
35			NBS (1.2)	0	30	45 (**10l**/**12l**)	80:20

[a] The ratios of **10**/**12** or **11**/**13** determined by the ^1H NMR analysis.

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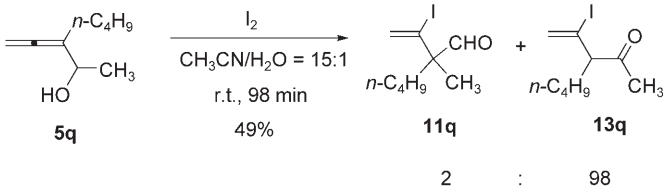
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CH_3CN (CH_3CN:H_2O = 15:1) afforded 3-bromo-2-(*n*-butyl)-2-phenyl-3-butenal **10a** in 77% yield.[12] By screening the most commonly used solvents and the ratio of MeCN/H_2O, it was found that a mixed solvent of MeCN/H_2O (15:1) is still the best reaction solvent.

Subsequently, the reaction of secondary 2,3-allenols with X_2 (or NXS) was performed under these conditions (Table 4). It can be concluded that 2,3-allenols with differently substituted phenyl groups at the 1-position may be used to prepare 3-halo-3-enals in good yields. By introducing the allyl group to the 2-position of 2,3-allenols, halogenation of the allylic C–C double bond was not observed, indicating a much higher reactivity of the allene functionality. For the synthesis of bromides, both Br_2 and NBS may be applied, however, the yields with NBS are usually higher than those with Br_2, probably due to the higher reactivity of Br_2 towards the remaining C=C bond (compare entries 2/3, 5/6, 12/13, 16/17, 23/24 and 30/31, Table 4). The corresponding reaction with I_2 can also occur smoothly to afford 3-iodo-3-enals **11** in good to high yields (entries 4, 7, 11, 14, 18 and 21, Table 4). The structures of the products were determined unambiguously by the X-ray diffraction study of **11b**.[12] Again for the same reason the yields with NIS are higher than those with I_2 (compare entries 7/8, 14/15, 21/22). The reaction of **5h** and **5i** with I_2 at room temperature afforded a complicated mixture (entries 25 and 28, Table 4), while the reaction of **5i** with I_2 at −20 °C afforded **11i** in 37% yield (entry 29, Table 4). In some cases, a maximum of 5% of β,γ-enone products **12** or **13** were also formed (entries 1, 9–11, 19–22, 29–31, Table 4). The ratio of **10**/**12** or **11**/**13** depends on the electronic effect of the substituent of the aryl group at the 1-position: with a more electron-donating group, the ratio of **10**/**12** or **11**/**13** is higher.

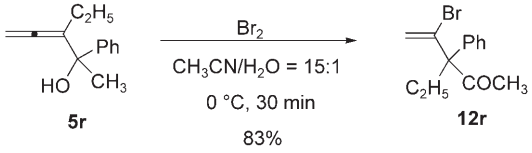
However, the substrates with a naphthyl or an aryl group bearing an electron- withdrawing group afforded a mixture of aldehydes **10**/**12** and ketones **11**/**13** with relatively poor selectivity (entries 32–35, Table 4). Again, it should be noted that the reactions of **5k–l** with "I^+" are quite complicated, failing to afford the expected iodides.

When the 1-aryl group was replaced with an alkyl group, i.e., a methyl group, a 2-halo-2-alkenyl ketone, i.e., **13q**, was formed as the major product (Scheme 3).



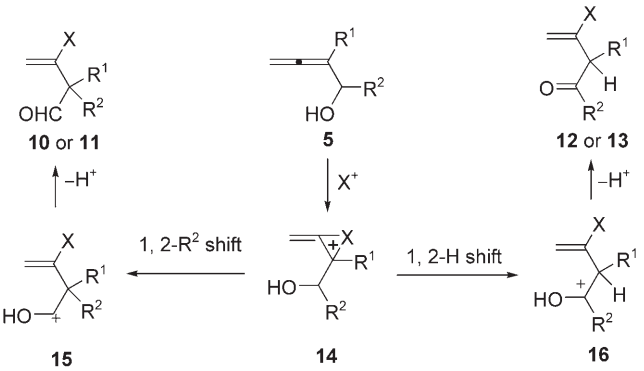
Scheme 3.

Then the reaction was extended to the tertiary 2,3-allenol, i.e., 2-phenyl-3-ethylpenta-3,4-dien-2-ol **5r** affording 4-bromo-4-enone **12r** in 83% yield (Scheme 4).



Scheme 4.

A rationale for this transformation is shown (Scheme 5). The interaction of "X^{+}" with the internal C=C bond of the allene moiety would form intermediate **14**, which would undergo a 1,2-shift of the Ar group to open the three-membered ring forming the cationic homoallylic alcohol intermediate **15**. Upon releasing of H^+ the reaction would form the final 3-halo-3-enals **10** (or **11**). If a 1,2-H transfer occurred, the reaction would form the ketone product **12** (or **13**) *via* the intermediacy of **16**.



Scheme 5.

Conclusions

In summary, it was observed that the reaction of primary 2,3-allenols with I_2 afforded 3-iodo-2,5-dihydrofurans. Under the similar reaction conditions, the reaction of 1-aryl-2,3-allenols afforded 3-halo-3-enals highly selectively when an electron-donating substituent is connected to the aryl group at the 1-position. However, with a relatively electron-withdrawing aryl group or a naphthyl group at the 1-position, a 1,2-H transfer was observed as the minor pathway as well; With an alkyl group group at the 1-position, the 1,2-H shift was observed as the major pathway to afford the 2-halo-2-alkenyl ketones as the major products. Due

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to the easy availability of the starting compounds **5** and the not readily available nature as well as the synthetic potential of 3-halo-3-enals or 2-halo-2-alkenyl ketones, this method will be very useful in organic synthesis.[12] Further studies are being carried out in our laboratory.

Experimental Section

General Methods

All the reactions were carried out in the open air unless otherwise indicated. ^{1}H and ^{13}C NMR spectra were recorded on a Bruker AM-300 spectrometer or Bruker AM-400 spectrometer. IR spectra were recorded on a Bruker Vector 22 spectrometer. Mass spectra were recorded on a HP 5989 A spectrometer. Flash-column chromatography was carried out on silica gel H (10–40 μ). Petroleum ether with a boiling point range from 60 to 90 °C was used. For the data of compounds reported in the previous communication,[12] see the related supporting information attached to this publication.

Primary 2,3-Allenols

Compounds **2a–i** were prepared according to the known procedure from 2,3-allenoate **1a–i**.[13]

2-Propyldeca-2,3-dien-1-ol (2d); Typical Procedure: DIBAL-H (51.9 mL, 1.0 mol/L solution in toluene, 59.1 mmol) was added dropwise to a solution of 5.88 g (24.7 mmol) of **1d** in 35 mL of anhydrous toluene at −78 °C. After 6.2 h at −78 °C the reaction was complete as monitored by TLC. The mixture was then quenched with 25 mL of methanol and washed with 300 mL of aqueous HCl solution (5%). The organic layer was separated and the aqueous layer was extracted with ether. The combined organic layer was dried over Na_2SO_4, filtered, and evaporated. The residue was purified by column chromatography on silica gel (eluent: petroleum ether/ethyl acetate = 5:1) to give **2d** as an oil; yield: 3.39 g (70%); ^{1}H NMR (300 MHz, $CDCl_3$): δ = 5.39–5.30 (m, 1H), 4.07–3.93 (m, 2H), 2.05–1.91 (m, 4H), 1.53–1.21 (m, 11H), 0.93 (t, J = 7.5 Hz, 3H), 0.88 (t, J = 7.2 Hz, 3H); ^{13}C NMR (75 MHz, $CDCl_3$): δ = 198.7, 105.4, 95.7, 63.0, 31.7, 31.4, 29.3, 28.8, 22.6, 20.9, 14.0, 13.8; IR (neat): ν = 3316, 1964, 1465, 1378, 1156, 1092, 1017 cm^{-1}; MS (70 eV, EI): m/z (%) = 196 (M^+, 0.13), 165 (M^+-CH_2OH, 1.79), 71 (100); elemental analysis: calcd for $C_{13}H_{24}O$ (%): C 79.53, H 12.32; found: C 79.51, H 12.36.

Synthesis of Secondary 2,3-Allenols

Compounds **5a–q** were prepared according to the known procedure[14]

2-Butyl-1-phenylbuta-2,3-dien-1-ol (5a); Typical Procedure: A solution of 1.43 g (8.2 mmol) of **3a**, 2.21 g (11.5 mmol) of $SnCl_2$, and 1.73 g (11.4 mmol) of NaI in 40 mL of DMF was stirred for 1 h at room temperature. Then a solution of 1.16 mL (11.4 mmol) of aldehyde **4a** in 40 mL of DMF was added dropwise at 0 °C. After 12 h at 0 °C as monitored by TLC, the mixture was quenched with 15 mL of water, extracted with diethyl ether, washed with saturated NaCl, and dried over anhydrous Na_2SO_4. Evaporation and column chromatography on silica gel (petroleum ether/ethyl acetate = 10:1) afforded **5a**[16] as an oil; yield: 0.66 g (40%); ^{1}H NMR (400 MHz, $CDCl_3$): δ = 7.40–7.26 (m, 5H), 5.10 (s, 1H), 5.04–4.96 (m, 2H), 2.21 (bs, 1H), 1.90–1.73 (m, 2H), 1.42–1.32 (m, 2H), 1.32–1.22 (m, 2H), 0.84 (t, J = 7.2 Hz, 3H).

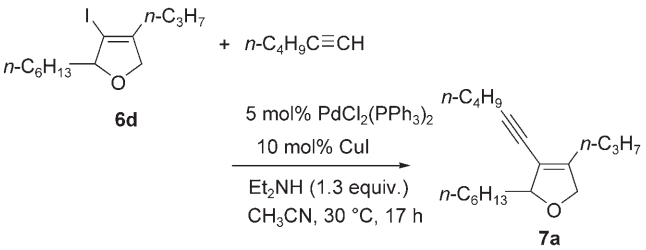
Cyclization of Primary 2,3-Allenols with I_2; General Procedure

A solution of **2** (0.5 mmol) and iodine (1.25 mmol) in 4 mL of MeCN/H_2O (15:1) was stirred at room temperature. When the reaction was complete as monitored by TLC, the mixture was then quenched with 6 mL of water, which was followed by the addition of a saturated aqueous solution of $Na_2S_2O_3$. This mixture was extracted with diethyl ether (3 × 25 mL), washed with an aqueous solution of NaCl, and dried over Na_2SO_4. Concentration and column chromatography on silica gel (petroleum ether/ethyl acetate = 40:1) afforded **6**.

2-Heptyl-3-iodo-2,5-dihydrofuran (6a): The reaction of 85.7 mg (0.5 mmol) of **2a** and 319.5 mg (1.25 mmol) of I_2 in 4 mL of MeCN and 0.27 mL of H_2O at rt for 80 min afforded **6a** as an oil; yield: 97.5 mg (65%); ^{1}H NMR (400 MHz, $CDCl_3$): δ = 6.21 (d, J = 1.6 Hz, 1H), 4.72–4.64 (m, 1H), 4.60–4.50 (m, 2H), 1.80–1.71 (m, 1H), 1.56–1.44 (m, 1H), 1.40–1.21 (m, 10H), 0.87 (t, J = 6.8 Hz, 3H); ^{13}C NMR (100 MHz, $CDCl_3$): δ = 134.8, 92.1, 89.8, 76.2, 33.8, 31.8, 29.5, 29.2, 24.1, 22.6, 14.1. IR (neat): ν = 1612, 1464, 1345, 1237, 1068, 1008 cm^{-1}; MS (70 ev, EI): m/z (%) = 294 (M^+, 2.73), 195 (100); HR-MS: m/z = 294.0479, calcd. for $C_{11}H_{19}IO$ (M^+): 294.0481.

Coupling Reactions of 3-Iododihydrofuran 6d

Sonogashira Coupling of 6d with 1-Hexyne to Afford 2-Hexyl-3-hexynyl-4-propyl-2,5-dihydrofuran (7a)



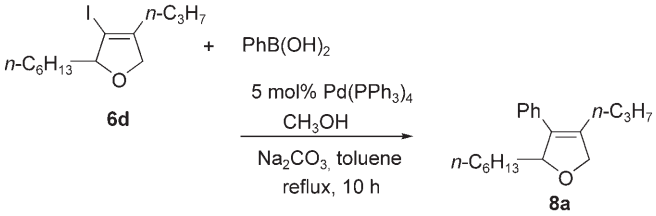
A mixture of 82.0 mg (0.25 mmol) of **6d**, 26.9 mg (0.325 mmol) of 1-hexyne, 24.7 mg (0.325 mmol) of Et_2NH, 4.0 mg (10 mol%) of CuI, and 8.5 mg (5 mol%) of $PdCl_2(PPh_3)_2$ in 2 mL of CH_3CN was stirred at 30 °C under N_2 for 17 h (monitored by TLC, petroleum ether/ethyl acetate = 40:1). Water (6 mL) was added and the reaction mixture was extracted with ether. The combined extract was washed with saturated brine. After drying over Na_2SO_4, filtration, and evaporation, the residue was purified by column chromatography on silica gel (petroleum ether/ethyl acetate = 100:1) to afford **7a** as an oil; yield: 67.3 mg (96%); ^{1}H NMR (400 MHz, $CDCl_3$): δ = 4.74–4.68 (m, 1H), 4.59–4.50 (m, 2H), 2.36 (t, J = 6.8, 2H), 2.25–2.20 (m, 2H), 1.75–1.67 (m,

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1H), 1.56–1.20 (m, 15H), 0.94–0.85 (m, 9H); ^{13}C NMR (100 MHz, $CDCl_3$): δ=143.9, 119.1, 96.1, 87.7, 76.5, 73.3, 34.9, 31.8, 30.9, 29.4, 28.8, 24.6, 22.6, 21.8, 21.0, 19.2, 14.1, 13.9, 13.5; IR (neat): ν=2221, 1653, 1464, 1432, 1379, 1258, 1061 cm^{-1}; MS (70 eV, EI): *m/z* (%)=276 (M^+, 10.67), 191 (100); HR-MS: *m/z*=276.2443; calcd. for $C_{19}H_{32}O$: 276.2453.

Suzuki Coupling of 6d with Phenylboronic Acid to Afford 2-Hexyl-3-phenyl-4-propyl-2,5-dihydrofuran (8a)

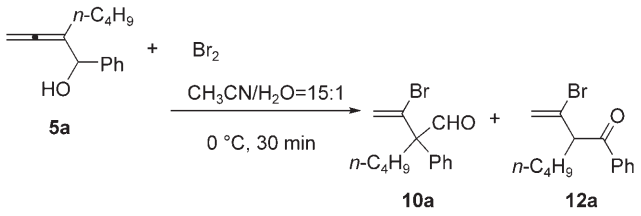


The mixture of 82.5 mg (0.25 mmol) of **6d**, 62.0 mg (0.5 mmol) of phenylboronic acid, 15.5 mg (5 mol%) of $Pd(PPh_3)_4$, 0.1 mL of CH_3OH, and 0.3 mL (2 M in H_2O) of Na_2CO_3 in 2 mL of toluene was refluxed under N_2 for 10 h as monitored by TLC (petroleum ether/ethyl acetate=40:1). Water (6 mL) was added and the reaction mixture was extracted with ether. The combined extract was washed with saturated NaCl. After drying over Na_2SO_4, filtration, and evaporation, the residue was purified by column chromatography on silica gel to afford **8a** as an oil; yield: 64.2 mg (92%); ^{1}H NMR (400 MHz, $CDCl_3$): δ=7.29 (t, *J*=7.2 Hz, 2H), 7.22–7.19 (m, 1H), 7.12 (d, *J*=8.0 Hz, 2H), 5.20–5.14 (m, 1H), 4.72 (dd, J_1=12.4, J_2=5.6 Hz, 1H), 4.64 (dd, J_1=12.4, J_2=2.8 Hz, 1H), 2.20–2.09 (m, 2H), 1.49–1.14 (m, 12H), 0.83 (t, *J*=7.2 Hz, 3H), 0.77 (t, *J*=7.2 Hz, 3H); ^{13}C NMR (100 MHz, $CDCl_3$): δ=135.05, 134.95, 134.8, 128.3, 128.2, 126.9, 88.8, 77.6, 34.6, 31.8, 29.3, 27.9, 24.7, 22.5, 21.5, 14.1, 14.0; IR (neat): ν=1719, 1597, 1490, 1464, 1378, 1065 cm^{-1}; MS (70 eV, EI): *m/z* (%)=272 (M^+, 9.67), 271 (M^+−H, 36.37), 187 (100); HR-MS: *m/z*=272.2144, calcd. for $C_{19}H_{28}O$: 272.2135.

Reaction of Secondary 2,3-Allenols (5) with X_2 (or NXS)

3-Bromo-2-butyl-2-phenyl-3-butenal (10a); Typical Procedure

To a solution of 82.4 mg (0.41 mmol) of **5a** in 2.4 mL of MeCN and 0.27 mL of H_2O was added with stirring 1.6 mL (0.5 M in MeCN, 0.8 mmol) of Br_2 at 0 °C for 30 min. After complete consumption of the allenol as monitored by TLC, the mixture was quenched with 6 mL of water followed by the addition of a saturated aqueous solution of $Na_2S_2O_3$, extracted with diethyl ether (25 mL×3), washed with NaCl, and dried over anhydrous Na_2SO_4. Evaporation and column chromatography on silica gel (petroleum ether/ethyl acetate=100:1) afforded a mixture of **10a**[12] and **12a**; yield: 91.2 mg (80%) (**10a**:**12a**≥95:5, determined by ^{1}H NMR).



The following data are discernible for **12a**: ^{1}H NMR: δ=5.78 (d, *J*=2.2 Hz, 1H), 5.63 (d, *J*=2.2 Hz, 1H).

3-Bromo-2-butyl-2-(1-naphthyl)-3-butenal (10k)

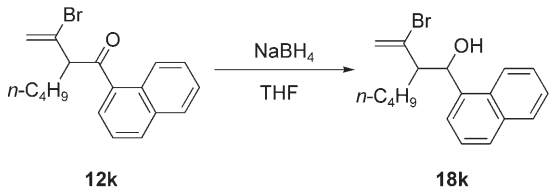
The reaction of 100.2 mg (0.4 mmol) of **5k** and 1.6 mL (0.5 M in MeCN, 0.8 mmol) of Br_2 in 2.4 mL of MeCN and 0.27 mL of H_2O at 0 °C for 30 min afforded a mixture of **10k** and **12k** (90:10); yield: 108.1 mg (73%).

10k: oil, ^{1}H NMR (300 MHz, $CDCl_3$): δ=9.86 (s, 1H), 8.15–8.09 (m, 1H), 7.95–7.83 (m, 2H), 7.55–7.41 (m, 4H), 6.26 (s, 1H), 6.04–5.95 (m, 1H), 2.55–2.41 (m, 1H), 2.35–2.20 (m, 1H), 1.50–1.23 (m, 3H), 0.89 (t, *J*=6.9 Hz, 4H); ^{13}C NMR (75 MHz, $CDCl_3$): δ=197.1, 134.5, 134.1, 133.3, 131.5, 129.5, 129.3, 126.5, 126.1, 125.7, 125.0, 124.9, 122.4, 64.9, 33.0, 27.1, 23.0, 13.9; IR (KBr): ν=3050, 2957, 2714, 1731, 1599, 1510, 1466, 1397, 1343, 1095, 1025 cm^{-1}; MS (70 eV, EI): *m/z* (%)=332 [M^+ (^{81}Br), 0.22], 330 [M^+ (^{79}Br), 0.23], 303 [M^+ (^{81}Br)−CHO, 5.24], 301 [M^+ (^{79}Br)−CHO, 4.93], 165 (100); HR-MS (EI): *m/z*=330.0626, calcd. for $C_{18}H_{19}{}^{79}BrO$: 330.0619.

The following data are discernible for **12k**: ^{1}H NMR: δ=5.82 (d, *J*=2.2 Hz, 1H), 5.62 (d, *J*=2.2 Hz, 1H).

The reaction of 75.5 mg (0.3 mmol) of **5k** and 64.2 mg (0.36 mmol) of NBS in 4 mL of MeCN and 0.27 mL of H_2O at 0 °C for 50 min afforded a mixture of **10k** and **12k** (93:7); yield: 75.8 mg (76%).

Reduction of Crude 3-Bromo-2-butyl-1-(1-naphthyl)-3-buten-1-one (12k) with $NaBH_4$; Synthesis of 3-Bromo-2-butyl-1-(1-naphthyl)-3-buten-1-ol (18k)



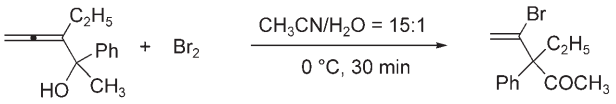
A solution of 38.3 mg of crude **12k** obtained from chromatographic separation and 3.5 mg (0.092 mmol) of $NaBH_4$ in 3 mL of THF was stirred at room temperature for 26 h (monitored by TLC, petroleum ether/ethyl acetate=5:1). The mixture was then quenched with 2 mL of water followed by extraction with diethyl ether (3×10 mL), washed with saturated brine, and dried over anhydrous Na_2SO_4. Evaporation and column chromatography on silica gel (petroleum ether/ethyl acetate=20:1) afforded **18k** as an oil;

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yield: 26.5 mg (90%); ^{1}H NMR (300 MHz, $CDCl_3$): $\delta = 8.29$ (d, $J = 8.1$ Hz, 1H), 7.89 (d, $J = 8.4$ Hz, 1H), 7.82 (d, $J = 8.1$ Hz, 1H), 7.61 (d, $J = 7.2$ Hz, 1H), 7.56–7.43 (m, 3H), 5.91 (s, 1H), 5.75 (s, 1H), 5.36 (d, $J = 9.3$ Hz, 1H), 2.86–2.74 (m, 1H), 2.28 (d, $J = 1.8$ Hz, 1H), 1.49–1.32 (m, 1H), 1.23–0.84 (m, 4H), 0.82–0.73 (m, 1H), 0.70 (t, $J = 6.9$ Hz, 3H); ^{13}C NMR (75 MHz, $CDCl_3$): $\delta = 137.3$, 136.7, 133.9, 131.5, 128.9, 128.6, 126.1, 125.6, 125.3, 123.8, 120.8, 77.4, 72.7, 57.3, 29.0, 28.9, 22.1, 13.8; IR (KBr): $\nu = 3446$, 3048, 2955, 2858, 1625, 1597, 1511, 1465, 1394, 1378, 1260, 1225, 1166, 1046 cm^{-1}; MS (70 eV, EI): m/z (%) = 334 [M^+ (^{81}Br), 0.13], 332 [M^+ (^{79}Br), 0.12], 317 [M^+ (^{81}Br)−OH, 1.25], 315 [M^+ (^{79}Br)−OH, 1.15], 157 (100); elemental analysis: calcd. for $C_{18}H_{21}BrO$ (%): C 64.87, H 6.35; found: C 64.83, H 6.37.

Reaction of Tertiary 2,3-Allenol (5r) with Br_2; Synthesis of 4-Bromo-3-ethyl-3-phenyl-4-penten-2-one (12r)



5r **12r**

The reaction of 56.4 mg (0.30 mmol) of **5r** and 0.72 mL (0.5 M in MeCN, 0.36 mmol) of Br_2 in 3.3 mL of MeCN and 0.27 mL of H_2O at 0 °C for 30 min afforded **12r** as an oil: yield: 66.2 mg (83%); ^{1}H NMR (300 MHz, $CDCl_3$): $\delta = 7.41$–7.27 (m, 5H), 6.07 (d, $J = 2.4$ Hz, 1H), 5.94 (d, $J = 2.4$ Hz, 1H), 2.29 (q, $J = 7.2$ Hz, 1H), 2.29 (q, $J = 7.2$ Hz, 1H), 2.10 (s, 3H), 0.91 (t, $J = 7.2$ Hz, 3H); ^{13}C NMR (100 MHz, $CDCl_3$): $\delta = 205.0$, 138.2, 135.2, 128.6, 128.4, 127.5, 122.4, 68.9, 27.3, 27.2, 9.3; IR (KBr): $\nu = 2975$, 2880, 1713, 1617, 1495, 1438, 1078 cm^{-1}; MS (70 eV, EI): m/z (%) = 268 [M^+ (^{81}Br), 0.05], 266 [M^+ (^{79}Br), 0.05], 239 [M^+ (^{81}Br)−C_2H_5, 2.39], 237 [M^+ (^{79}Br)−C_2H_5, 2.38], 129 (100); HR-MS: m/z = 291.0174, calcd. for $C_{13}H_{15}{}^{81}BrONa$ (M^++Na): 291.0184, 289.0192, calcd for $C_{13}H_{15}{}^{79}BrONa$ (M^++Na): 289.0204.

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